

EMCH 260

SOLID MECHANICS

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Torsion

5

Chapter 5.4

- ✓ Solve complex statically indeterminate torque-loaded problems with other constraints

PROCEDURE FOR ANALYSIS

Step 1: Equilibrium (exactly same as what we did before)

- Draw a free-body diagram of the shaft to identify torques acting on it
- If unknown reactions on free-body diagram greater than number of moment equilibrium equations, then problem is statically indeterminate
- Write the equations of moment equilibrium for the shaft

Step 2: Compatibility

- Express compatibility conditions in terms of twist angles caused by torques

Step 3: Load-Displacement

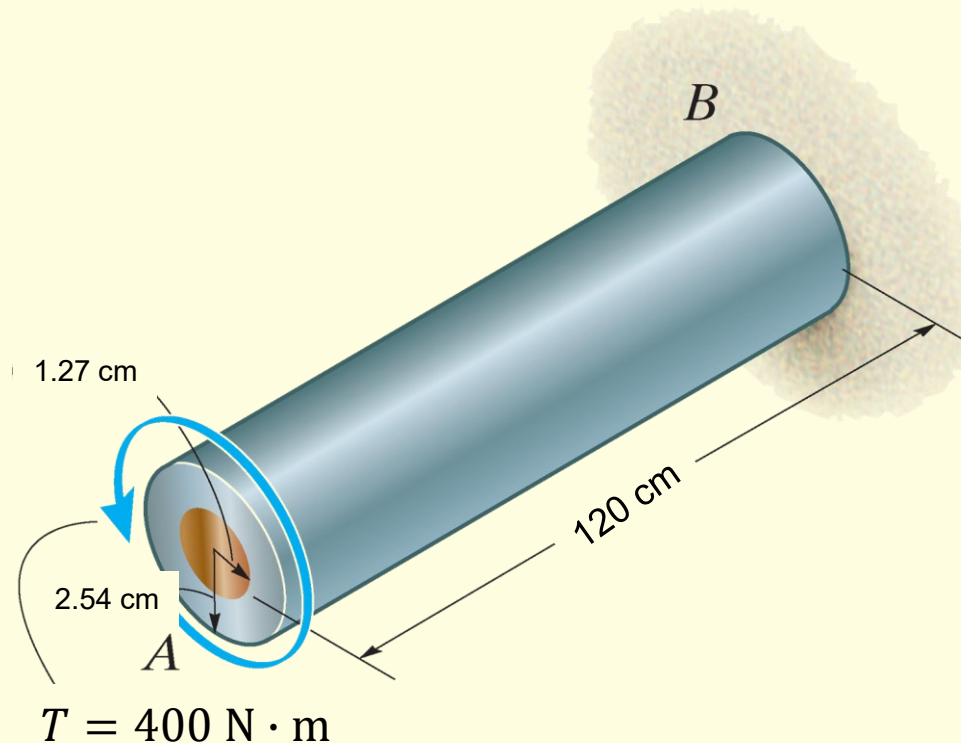
- Use load-displacement relations (such as $\phi = TL/(JG)$) to relate unknown twist angles to reactions
- Solve the equations for the unknown reactive torques. If any of the magnitudes is negative, it means the torque acts in opposite direction of that indicated on free-body diagram.

Step 4: Other Constrains

- Apply any other constrains to generate additional equation(s)

IN-CLASS EXAMPLE

The shaft shown is made from a steel tube, which is bonded to a brass core. If a torque of $400 \text{ N} \cdot \text{m}$ is applied to its end, determine the internal torques and plot the shear-stress distribution along a radial line on its cross section. Take $G_{\text{st}} = 78.6 \text{ GPa}$, $G_{\text{br}} = 35.8 \text{ GPa}$.



IN-CLASS EXAMPLE (cont'd)

Solutions

Step 1: Equilibrium

Assume internal resultant torques resisted by the steel and brass as T_{st} and T_{br} , respectively, and apply free body diagram to the entire shaft:

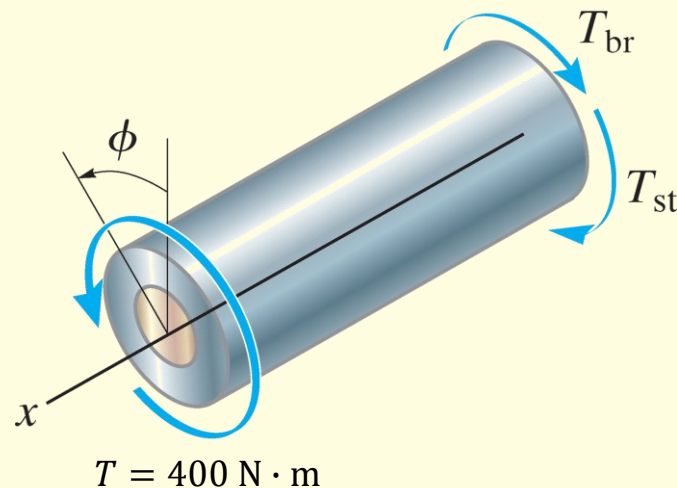
$$T_{st} + T_{br} = T = 400 \quad (1)$$

Step 2: Compatibility

The steel tube and the brass core are bonded together, meaning that the angle of twist of end A to be the same for both the steel and brass.

Thus,

$$\phi = \phi_{st} = \phi_{br} \quad (2)$$



IN-CLASS EXAMPLE (cont'd)

Solutions

Step 3: Load-Displacement

Applying the load-displacement relationship (angle of twist equation) for the steel and brass separately:

$$\phi_{st} = \frac{T_{st} \cdot L}{J_{st} \cdot G_{st}} \quad (3a)$$

$$\phi_{br} = \frac{T_{br} \cdot L}{J_{br} \cdot G_{br}} \quad (3b)$$

Substituting Eqs. (3a) and (3b) into Eq. (2) we have:

$$\frac{T_{st} \cdot L}{J_{st} \cdot G_{st}} = \frac{T_{br} \cdot L}{J_{br} \cdot G_{br}}$$

$$\frac{T_{st}}{\frac{\pi}{2} (0.0254^4 - 0.0127^4) \times 78.6 \times 10^9} = \frac{T_{br}}{\frac{\pi}{2} \times 0.0127^4 \times 35.8 \times 10^9}$$

$$T_{st} = 32.93 T_{br} \quad (3c)$$

IN-CLASS EXAMPLE (cont'd)

Solutions

Substituting Eq. (3c) into Eq. (1) we have:

$$T_{st} = 388.2 \text{ N} \cdot \text{m}$$

$$T_{br} = 11.8 \text{ N} \cdot \text{m}$$

Step 3: Shear-Stress Distribution

The shear stress in the brass core varies linearly from zero at its center to a maximum at the interface where it contacts the steel tube. Using the torsional formula

$$(\tau_{br})_{\max} = \frac{T_{br} c_{br}}{J_{br}} = \frac{11.8 \times 0.0127}{\frac{\pi}{2} \times 0.0127^4} = 3.67 \text{ MPa}$$

For the steel, the minimum and maximum shear stress are

$$(\tau_{st})_{\min} = \frac{T_{st} c_1}{J_{st}} = \frac{388.2 \times 0.0127}{\frac{\pi}{2} (0.0254^4 - 0.0127^4)} = 8.04 \text{ MPa}$$

$$(\tau_{st})_{\max} = \frac{T_{st} c_2}{J_{st}} = \frac{388.2 \times 0.0254}{\frac{\pi}{2} (0.0254^4 - 0.0127^4)} = 16.08 \text{ MPa}$$

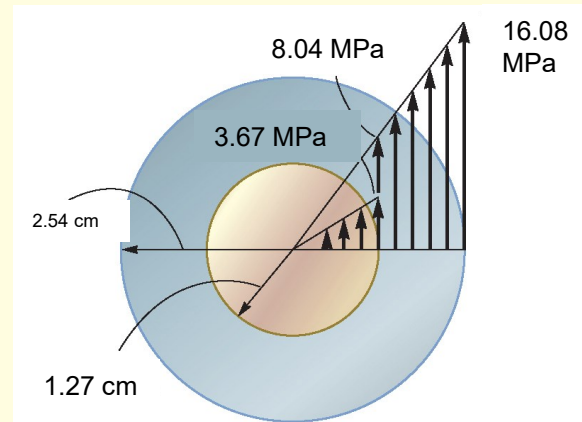
IN-CLASS EXAMPLE (cont'd)

Solutions

The results are plotted in the right figures.

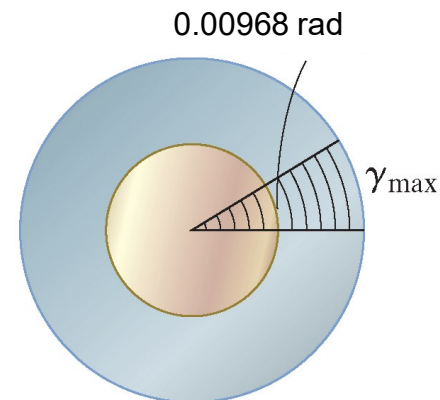
Note the shear stress at the brass and steel interface is discontinuous. This is expected, since the materials have different moduli of rigidity, i.e. steel is stiffer than brass, and thus it carries more shear stress at the interface.

Although the shear stress is discontinuous, the shear strain is not. Rather, it is the same on either side of the brass-steel interface, which is guaranteed by the compatibility condition applied (the same angle of twist for the steel and brass).



Shear-stress distribution

(c)



Shear-strain distribution

(d)